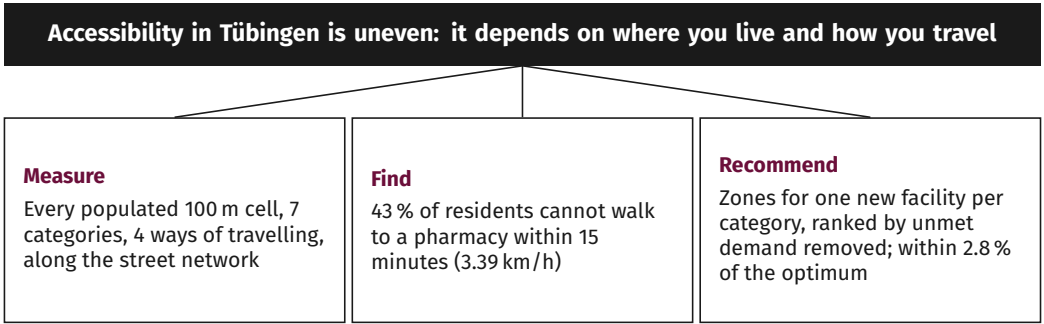
reach: 15-minute accessibility for every resident of Tübingen

Interim presentation · DS500 Data Science Project, Topic 9 “The 15-Minute City”

Rico · Socke · Juju
University of Tübingen, winter term 2026/27
25 November 2026

reach shows who cannot reach everyday services within 15 minutes, and where a new facility would help most



Automated from open data with one command; 9 of 11 validation experiments done

Source: run 20261110T223016Z (OSM 8 Nov 2026); validation report, run 20261106T142630Z.

The 15-minute city asks whether residents reach daily needs without a car; we make that question measurable and actionable for Tübingen

The idea

- Everyday services within a short walk or ride from home (Moreno et al., 2021)
- Mouratidis (2026): count *disaggregated* facilities per category and classify access as lack, insufficient or sufficient
- Applied once, largely by hand, to Copenhagen

What we add

- **Automation:** open data in, validated results out, with one command
- **Tübingen** with German open data (OSM, NVBW GTFS, Zensus 2022)
- **Inclusive profiles:** slow walking, wheelchair, cycling
- **A recommender:** where would one new facility help most?

Source: Moreno et al. (2021); Mouratidis (2026), full references in the backup.

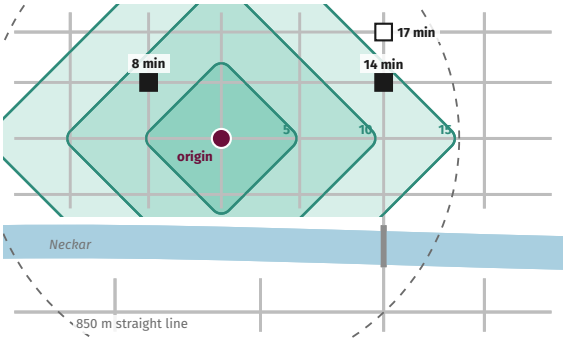
An isochrone shows every place you can reach from one point within a fixed time

Isochrone

$$I(o,t) = \{\, x \;:\; \tau_m(o,x) \le t \,\}$$

o: origin, t: time budget, τ_m: travel time along the street network for profile m

- Our budget: $t = 15$ min
- At 3.39 km/h: $\approx$ 850 m *along streets*
- Follows the network, stops at barriers like the Neckar



Schematic. Filled squares: reachable within 15 min; contours at 5, 10 and 15 min.

Source: methodology.md §1 (profiles); isochrones are computed by Valhalla.

We measure all 1,699 populated 100 m cells (90,355 residents) for seven categories and four ways of travelling

Who: Zensus 2022

Residents per 100 m grid cell; the cell centroid is the origin

What: OpenStreetMap

Street network and facilities: grocery, pharmacy, doctors, childcare, school, parks, playgrounds

Transit: NVBW GTFS

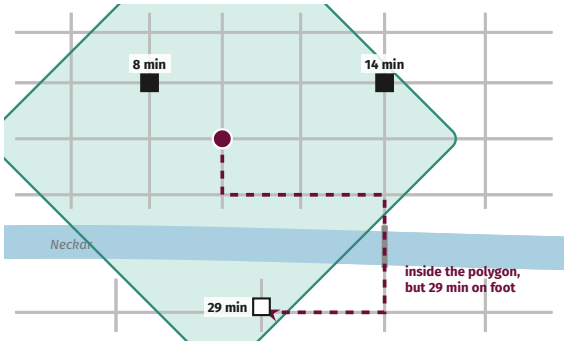
Weekday departures at stops within a 10-minute walk

Profile	Speed	Reach in 15 min
Walking (inclusive default)	3.39 km/h	850 m
Walking (comparison)	5.0 km/h	1,250 m
Wheelchair	3.39 km/h	850 m
Cycling	15 km/h	3,750 m

Reach = straight-line limit; the network time decides. Slopes slow walkers and wheelchairs down (Valhalla's slope model).

Source: run 20261110T223016Z (OSM 8 Nov 2026); methodology.md §1–§3.

We count facilities along the network, because isochrone polygons would add 18 % unreachable cell–facility pairs



Matrix, not polygon: for every cell, one Valhalla time matrix to all candidate facilities; a facility counts if its travel time $\leq$ 15 min.

Test V1 (150 cells, walking 3.39 km/h):

- Polygons add 18 % unreachable pairs and miss almost none
- 47 % of these lie across the Neckar, rail lines or main roads
- 5 % of classes would change

Source: validation report V1, run 20261106T142630Z; architecture.md §6.3 (decision D2).

Each cell gets a class per category, and a deficit that weighs missing access by residents

$A_{i,c}$ = facilities of category c reachable from cell i in 15 min, k_c = facilities needed (grocery, doctors, childcare: 2; others: 1)

Lack	$A_{i,c} = 0$
Insufficient	$0 < A_{i,c} < k_c$
Sufficient	$A_{i,c} \geq k_c$

Deficit (unmet demand, residents P_i)

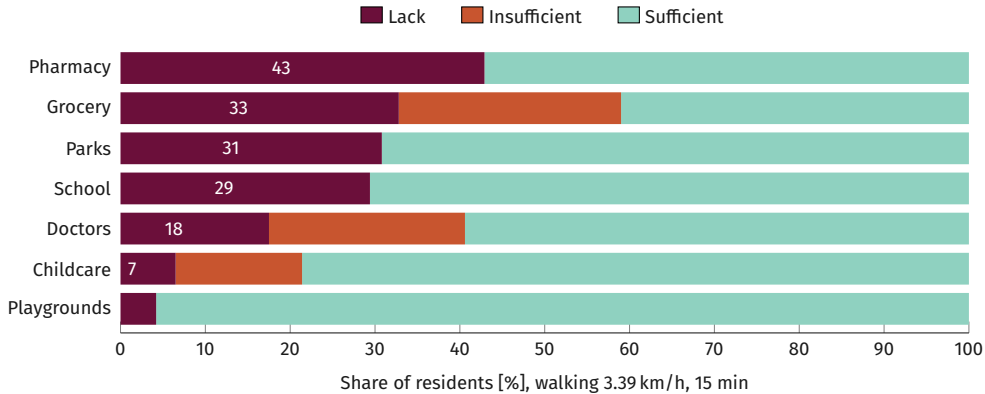
$$U_{i,c} = P_i \cdot \max\left(0,\ 1 - \frac{A_{i,c}}{k_c}\right)$$

Example: 120 residents, one supermarket ($k = 2$) $\Rightarrow U = 60$.

We store only raw counts and times; classes and deficits are derived when read, so a new k_c needs no re-routing.

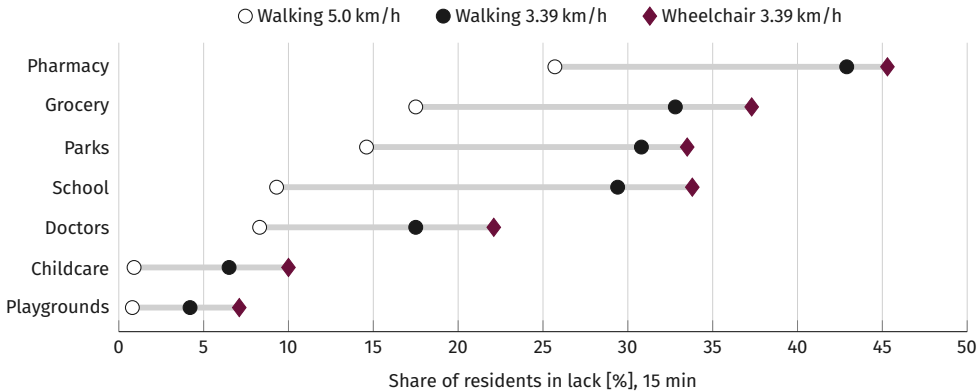
Source: methodology.md §2, §4.

More than four in ten residents cannot walk to a pharmacy within 15 minutes



Source: run 20261110T223016Z (OSM 8 Nov 2026), 90,355 residents in 1,699 cells. Insufficient exists only where $k_C = 2$.

Faster walkers reach far more; pharmacy and grocery stay the weakest on foot and by wheelchair



Source: run 20261110T223016Z (OSM 8 Nov 2026). Cycling 15 km/h: at most 3 % lack in every category. District rankings stay similar under sensitivity changes (V5).

A greedy recommender picks the zone that removes the most unmet demand, close to the exact optimum

Gain of a new facility of category c at zone x:

$$\Delta W(x,c) = \sum_{i \in C(x)} P_i \left[\min\left(1, \frac{A_{i,c}+1}{k_c}\right) - \min\left(1, \frac{A_{i,c}}{k_c}\right) \right]$$

1. Candidates: 100 m zones in the built-up area
2. Pick the largest ΔW, update the counts, repeat
3. Submodular objective: greedy reaches $\geq 1 - 1/e$ of the optimum

$\leq$ 2.8 %

below the exact MILP optimum
(90 instances, mean 0.17 %)

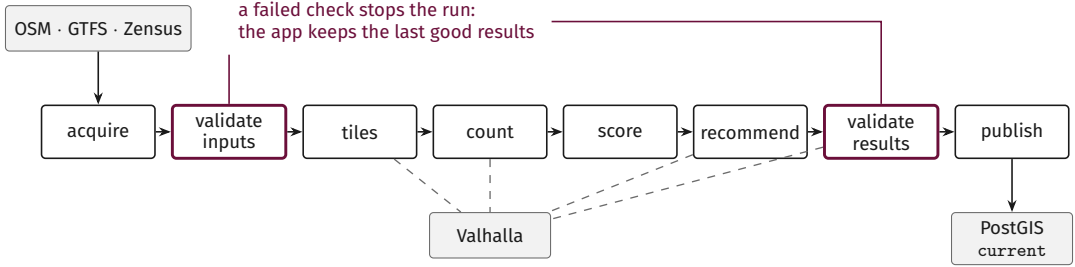
80–98 %

of picks near a removed facility when only the removed deficit counts
(back-test; random: 8–36 %)

Accessibility only: zoning, parcels and capacity are out of scope; results name zones, not sites.

Source: methodology.md §5; validation report V3, V4, run 20261106T142630Z.

One command turns open data into validated, published results



Reproducible: every run keeps its inputs, SHA-256 hashes and a manifest per stage

One routing engine: Valhalla for counting, isochrones and live queries in the app

Fast: the full count for Tübingen takes about 3.5 min on a laptop

Source: architecture.md §6–§7; `make refresh`.

Eleven experiments test the method, and the main design choices hold

18 %
V1: over-count avoided by matrix counting

95 %
V2: OSM recall for schools (childcare 88 %)

1.9 %
V6: class changes with building-based origins instead of centroids

0.5 s
V10: median difference matrix vs. single routes

- Deficits are spatially clustered (V8, Moran's I 0.72–0.88 on cells)
- Slopes change 3.3–3.6 % of classes, mostly on the hillsides (V11)
- Still open: plausibility with local knowledge (V7), reproducibility by a teammate (V9)

Source: validation report V1–V11, run 20261106T142630Z.

Live demo: anyone can click a place and see what is missing within 15 minutes

1. Class map: pharmacy, walking 3.39 km/h
2. Pin drop in an underserved area: what is reachable, what is missing, how long
3. Recommended zone: which residents would gain access

If the demo fails

We show screenshots of the same three steps instead.

By the submission on 22 January we close the open validations and host the app

Validation

- V7: local-knowledge check of the worst areas
- V9: a teammate reruns everything from the README
- Map the missing Kitas and schools in OSM (V2)

Product

- Hosted app and read-only database access for supervisors
- Weekly automatic refresh on the server
- Calibrate the transit supply classes to Tübingen

Known limit: the wheelchair profile avoids steps but not steep ramps, so wheelchair access on hillsides is optimistic.

Source: docs/notes.md N11, N13; validation report.

Thank you. Questions?

Accessibility in Tübingen is uneven: it depends on where you live and how you travel. reach measures it for every resident and shows where a new facility helps most.

Backup: references and data sources

- Moreno, C., Allam, Z., Chabaud, D., Gall, C., & Pratlong, F. (2021). Introducing the “15-Minute City”: Sustainability, resilience and place identity in future post-pandemic cities. *Smart Cities*, 4(1), 93–111.
- Mouratidis, K. (2026). *Travel Behaviour and Society*, 42, 101151.

Source	Used for	Licence
OpenStreetMap (Geofabrik extract)	Streets, facilities	ODbL 1.0
NVBW GTFS (naldo)	Stops, departures	dl-de/by-2-0, ODbL
Destatis Zensus 2022, 100 m grid	Residents per cell	dl-de/by-2-0
Valhalla (open source)	Routing, matrix, isochrones	MIT

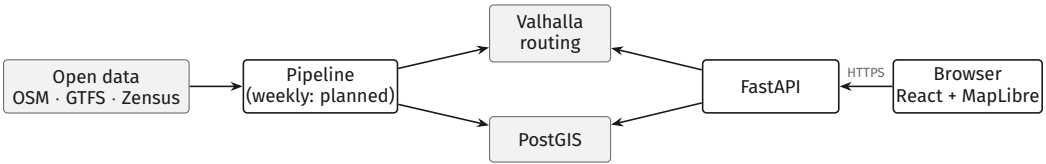
Source: README.md (data sources); Valhalla: licence file of the project repository.

Backup: validation experiments V1–V11

#	Experiment	Key result
V1	Matrix vs. isochrone polygons	Polygons add 18–20 % false pairs, 47 % across barriers
V2	OSM completeness	Childcare recall 88 %, precision 98 %; schools 95 % / 95 %
V3	Recommender back-test	80–98 % of picks within 500 m of a removed facility (only the removed deficit counts)
V4	Greedy vs. exact MILP	At most 2.8 % below the optimum (mean 0.17 %)
V5	Sensitivity (k, speed, budget)	Shares move strongly; district rankings stable (median Spearman 0.78–0.95)
V6	Centroid vs. building origin	1.9 % (walking) / 2.8 % (wheelchair) change class
V7	Local knowledge	Not yet run
V8	Spatial statistics	Moran's I 0.72–0.88 on cells; none between districts
V9	Reproducibility	Not yet run
V10	Matrix vs. single routes	Median $\lvert\Delta\rvert$ 0.5 s; 0 of 900 false negatives
V11	With vs. without slopes	3.3–3.6 % change class, mostly on hillsides

Source: docs/validation_report.md, run 20261106T142630Z.

Backup: architecture at runtime



Docker Compose on one small server; Caddy serves the web app over HTTPS. The API reads only the `current` views of the last validated run.

Source: architecture.md §3.